CYBER THREAT MONITORING

LEVEL 1 - GUIDED CAPSTONE

Operation Night Ledger

A five-day, team-based blue-team project: build a monitored Windows lab, connect it to the Wazuh SIEM, investigate a simulated intrusion, and report your findings like a professional Security Operations Center.

Duration	5 days (Day 1 build -> Day 5 present)
Format	Group project, worked online from one shared folder
Tools	VirtualBox, Windows + Event Viewer, Sysmon, Wazuh
You play	The BlueHarbor Finance SOC (blue team)
All files	CAPSTONE_RESOURCES (this folder)

Start here: read 'How to use this guide', then begin Day 1.
Safety: run all scripts only inside your lab virtual machines.

Team members: _____ Cohort: _____

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This guide is also a checklist: each day ends with a highlighted checkpoint box. Tick every item before moving on.

How to use this guide

Welcome to your **Cyber Threat Monitoring Level 1 Capstone**. Over five days your team will build a small security lab, switch on Windows logging, connect it to the **Wazuh** SIEM, investigate a simulated intrusion, and present your findings like a real Security Operations Center (SOC).

This is a **guided** capstone. Follow the days in order. Every step tells you exactly what to click or type, and every day ends with a **checkpoint** you can tick off.

> **One folder for everything.** All files you need are in the `CAPSTONE_RESOURCES` folder that came with this guide. Upload that one folder to your team's shared drive (Google Drive, OneDrive, or Teams). Nobody needs anything else.

> **Safety first.** Run every script and command only **inside your lab virtual machines** - never on your personal or work computer. The Day 4 "attack" is a harmless simulation. No real malware is used.

Legend used in this guide

DO THIS	A numbered action to perform.
CHECK	How to confirm the step worked.
TIP	Helpful advice or a shortcut.
WARNING	Something that can break your lab - read it.
FILE	A file to use from CAPSTONE_RESOURCES.

1. Capstone overview

What you will build

A two-machine lab inside **VirtualBox**:

- **CTM-WIN01** - a Windows workstation (the "victim"), with Windows auditing, **Sysmon**, and a **Wazuh agent**.
- **CTM-Wazuh** - the Wazuh all-in-one server (manager + indexer + dashboard), which collects and analyses the logs.

Learning outcomes

By the end, every team member can:

- Stand up a monitored Windows endpoint in a virtual lab.
- Enable and read the Windows logs a SOC relies on (Event Viewer).
- Collect endpoint logs centrally in a SIEM and write custom detection rules.
- Triage alerts, confirm them with raw evidence, and build an attack timeline.
- Map activity to **MITRE ATT&CK** and write an incident report.

Tools you will use (all free)

Tool	Role in the capstone
Oracle VirtualBox	Runs the virtual machines
Windows 11 / Server 2022 (eval)	The monitored endpoint
Windows Event Viewer	Read raw Windows logs (evidence)
Sysmon (Sysinternals)	Deep endpoint telemetry
Wazuh	SIEM - central collection, rules, dashboard
MITRE ATT&CK	Framework to classify attacker behaviour

The five days at a glance

Day	Theme	You finish with
Day 1	Build the lab	Two VMs on one network, snapshots taken
Day 2	Windows logging	Auditing + Sysmon on; Event Viewer views ready
Day 3	Wazuh SIEM	Agent ACTIVE; custom rules; logs flowing
Day 4	Attack & investigate	Triaged alerts; attack timeline; MITRE map
Day 5	Report & present	Incident report + team presentation

Team roles

Pick roles on Day 1 and record them in `07_Team_Templates / 01_Team_Charter.docx`. Teams of 3 combine roles; everyone investigates on Day 4.

Role	Owns
SOC Lead / Incident Commander	Stand-ups, task assignment, final timeline and decisions
Lab Engineer (Red Cell)	Builds and snapshots VMs; runs the Day 4 scenario
Windows Analyst	Event Viewer evidence and Windows Event IDs
Wazuh Analyst	Dashboard triage, custom rules, MITRE mapping
Scribe / Report Lead	Worksheet, incident report, presentation

2. The scenario: Operation Night Ledger

BlueHarbor Finance is a small (fictional) accounting firm. Your team is its SOC.

> At 07:45 this morning the Finance Manager reported that the file `Vendor_Bank_Details.csv` is missing from the shared finance workstation, and that "the computer felt slow last night." Helpdesk staff member **Jamie Santos (j.santos)** says they did not log in yesterday evening.

Management wants answers: Did someone break in, and how? What did they do? Was data taken or changed? Can they get back in? What must we fix? You will answer these on Days 4-5.

Day 1 - Build the lab

Goal: two virtual machines running and able to talk to each other on a private network. **Checkpoint:** you can ping between the VMs and open the Wazuh dashboard.

1.1 Download everything (do this first)

FILE

`02_Lab_Setup / Download_Links.md` has every link. Downloads total about 12 GB, so start them before anything else.

DO THIS

1. Download and install **VirtualBox 7.x** for your computer.
2. Download the **Wazuh OVA** (the ready-made SIEM virtual machine, ~5 GB).
3. Download a **Windows ISO**: Windows 11 Enterprise Evaluation (90 days). If your PC cannot run Windows 11, use Windows Server 2022 Evaluation instead.

> While downloads run, read `02_Lab_Setup / Lab_Network_Plan.md` and fill in the VM specs in your Team Charter.

1.2 Check your computer can run VMs

DO THIS

1. Reboot into your PC's BIOS/UEFI and make sure **virtualization** (Intel VT-x / AMD-V) is **enabled**. Without it, 64-bit VMs will not start.
2. Make sure you have at least **8 GB RAM** and **100 GB free disk**.

> If a member's laptop is too small, the team uses **one** strong PC as the "primary lab" and shares its screen during team calls. Decide this today.

1.3 Import the Wazuh server VM

DO THIS

1. VirtualBox menu: **File > Import Appliance**. Choose the Wazuh `.ova`.
2. Set RAM to **4096 MB** (8192 if you can), 2-4 CPUs. Name it **CTM-Wazuh**. Click Import.
3. Start the VM. At the login prompt sign in with **wazuh-user** / **wazuh**.
4. Type `ip -4 addr` and note the address (you will set this up in step 1.5).

1.4 Create the Windows endpoint VM

DO THIS

1. **Machine > New**. Name **CTM-WIN01**, Type Microsoft Windows, Version **Windows 11 (64-bit)**. VirtualBox 7 enables EFI, Secure Boot and TPM for you.
2. RAM **4096 MB**, CPUs **2**, disk **64 GB**. Attach the Windows ISO.
3. Install Windows. Create a local admin user named **ctmadmin** with a password you choose (write it

in your Team Charter). Set the computer name to **WIN-CTM-01**.

> To skip the "Microsoft account" requirement during Windows 11 setup, choose the **Windows Server** option instead, or select a limited-setup/"domain join" path when offered. A local account is all the lab needs.

1.5 Put both VMs on one private network

You need a **Host-only** network so the two VMs can talk privately, plus **NAT** so each can reach the internet for updates.

DO THIS

1. VirtualBox: **Tools > Network > Host-only Networks > Create**. Leave the default `192.168.56.0/24` with DHCP enabled.
2. For **each** VM: **Settings > Network**. Adapter 1 = **NAT**. Adapter 2 = **Host-only Adapter** (select the one you just created). Enable both.
3. Start both VMs. Find each host-only IP (starts with 192.168.56): on Wazuh type `ip -4 addr`; on Windows open Command Prompt and type `ipconfig`.
4. Record both IPs in your Team Charter.

CHECK

1. From the Windows VM, open a browser and go to `https://<WAZUH-IP>`. Accept the certificate warning. Log in with **admin** / **admin**. You should see the Wazuh dashboard. In the dashboard menu, open **About** and record the **Wazuh version** - you need it on Day 3.

1.6 Take your first snapshots

> Snapshots are your safety net. If anything breaks later, you roll back in seconds instead of rebuilding.

DO THIS

1. For each VM: VirtualBox **> (select VM) > Snapshots > Take**.
2. Name them **Day1-Clean-Build**.

> **Day 1 checkpoint - all must be true**
>
> ```
> [ ] VirtualBox installed, virtualization enabled
> [ ] CTM-Wazuh and CTM-WIN01 both boot
> [ ] Both VMs have a 192.168.56.x address
> [ ] Wazuh dashboard opens from the Windows VM; version recorded
> [ ] "Day1-Clean-Build" snapshot taken on both VMs
> [ ] Team Charter filled in (roles, IPs, version)
> ```

Day 2 - Turn on Windows logging

Goal: make the Windows endpoint record the security events a SOC needs. **Checkpoint:** `Preflight_Check.ps1` shows the logging items as PASS.

> **Running the scripts.** All Day 2-4 scripts run inside the Windows VM. Right-click **Windows PowerShell > Run as administrator**. If scripts are blocked, run this once in that same window:
> `Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass`
> Then run a script with `.\ScriptName.ps1`. Copy the CAPSTONE_RESOURCES folder into the VM first (shared folder, or just download it from your shared drive inside the VM).

2.1 Create the company environment

FILE

`03_Windows_Logging / Seed_Lab_Environment.ps1`

This creates three staff accounts (**j.santos**, **m.reyes**, **a.cruz**) and the `C:\CapstoneFinance` folder with sample business files. Note: j.santos is given a deliberately **weak password** - that is part of the story.

DO THIS - run it, then confirm the users appear with `Get-LocalUser`.

2.2 Enable Windows audit logging

FILE

`03_Windows_Logging / Enable_Audit_Logging.ps1`

This turns on auditing for logons, account changes, process creation and more; adds the **command line** to process events; enables **PowerShell script block logging**; and makes the logs bigger so evidence is not overwritten.

DO THIS - run it. It prints the audit policy at the end; screenshot that.

> After this runs, these events start recording: 4624/4625 (logon success/failure), 4720/4732 (account created / added to group), 4688 (process with command line), 4698 (scheduled task), 4104 (PowerShell code). See the cheat sheet below.

2.3 Install Sysmon

FILE

`03_Windows_Logging / Install_Sysmon.ps1` and `sysmon-capstone.xml`

Sysmon adds deep detail: every process (with hashes and parent), network connections, file creation/deletion, registry autostart changes and DNS queries. The script downloads Sysmon from Microsoft and installs it with our tuned configuration.

DO THIS - run it.

CHECK - open Event Viewer > Applications and Services Logs > Microsoft > Windows > Sysmon > Operational. You should see events appearing.

2.4 Import the ready-made Event Viewer views

FILE

`03_Windows_Logging / EventViewer_Custom_Views\` (four XML files)

These give you one-click filtered views instead of scrolling through thousands of events.

DO THIS

1. Open **Event Viewer** (press Start, type `eventvwr.msc`).
2. Right-click **Custom Views > Import Custom View...**
3. Import each of the four files: **CV1 Authentication**, **CV2 Account Changes**, **CV3 Persistence and Execution**, **CV4 PowerShell and Defender**.

2.5 Learn the key Event IDs

Keep `03_Windows_Logging / Windows_Event_ID_Cheat_Sheet.md` open while you work. The essentials:

Event ID	Meaning	Why it matters
4625	Failed logon	Password guessing / brute force
4624	Successful logon	Who got in, and how (Logon Type)
4720	User account created	Attacker making a backdoor account
4732	Added to a local group	Privilege escalation (e.g. Administrators)
4688	New process	What was run (with command line)
4698	Scheduled task created	Persistence
4104	PowerShell script block	Shows the actual code - even decoded
1102	Security log cleared	Always suspicious

> **Day 2 checkpoint**
> `[ ] Seed users and C:\CapstoneFinance created`
> `[ ] Audit logging enabled (screenshot of auditpol output)`
> `[ ] Sysmon service running and logging events`
> `[ ] Four custom views imported into Event Viewer`
> `[ ] Preflight_Check.ps1 shows logging items PASS`
> `[ ] Snapshot taken: "Day2-Logging-On"`

Day 3 - Send the logs to Wazuh (SIEM)

Goal: the Windows endpoint ships its logs to Wazuh, and your custom rules raise alerts. **Checkpoint:** the agent shows **ACTIVE** and your test events create alerts.

3.1 Install the Wazuh agent on Windows

FILE

`04_Wazuh / Install_Wazuh_Agent.ps1`

This installs the agent, points it at your Wazuh server, and adds the extra log sources (Sysmon,

PowerShell, Defender) plus **File Integrity Monitoring** on the finance folder.

> The agent version must be the **same or older** than your Wazuh server version (the one you recorded on Day 1). Pass it with `-Version`.

DO THIS - in the Windows VM, run (use your own IP and version):

```
.\Install_Wazuh_Agent.ps1 -WazuhIP 192.168.56.101 -Version 4.9.2
```

CHECK

1. The script prints the agent log; look for **"Connected to the server"**.
2. In the Wazuh dashboard open **Agents**. **WIN-CTM-01** should be **ACTIVE** (green).

> If the agent is not active: check the IP, confirm both VMs are on the host-only network, and that ports 1514/1515 are reachable (run `Preflight_Check.ps1 -WazuhIP <ip>`). See Troubleshooting.

3.2 Add the custom detection rules

FILE

`04_Wazuh / local_rules.xml`

These rules (IDs 100200-100210) are the brain of your SOC. They turn raw events into meaningful alerts: brute force, new admin accounts, scheduled tasks, encoded PowerShell, registry persistence, finance-file tampering, and Defender detections - each mapped to MITRE.

DO THIS

1. In the dashboard: **menu > Server management > Rules**.
2. Click **Manage rules files**, then **Custom rules**, and edit **local_rules.xml**.
3. Keep anything already there. Paste the contents of our `local_rules.xml` below it. Click **Save**.
4. When prompted, **Restart** the manager (or on the server console run `sudo systemctl restart wazuh-manager`).

> Alternative on the server console: edit `/var/ossec/etc/rules/local_rules.xml`, then restart the manager. After a restart, check `sudo tail -n 50 /var/ossec/logs/ossec.log` for rule errors.

3.3 Prove it works (smoke test)

DO THIS - in the Windows VM, generate a couple of safe test events:

1. Fail a logon on purpose a few times: open PowerShell and run `runas /user:WIN-CTM-01\j.santos cmd` and type a wrong password 3-4 times.
2. Create and remove a test file in the watched folder: `"test" | Out-File C:\CapstoneFinance\test.txt ; Remove-Item C:\CapstoneFinance\test.txt`

CHECK - in the dashboard go to Threat Hunting (or Discover) and search `rule.groups:"ctm_capstone"`. You should see your alerts. Use the queries in `04_Wazuh / Wazuh_Dashboard_Queries.md`.

Day 3 checkpoint

```
[ ] Wazuh agent ACTIVE in the dashboard
[ ] Sysmon / PowerShell / Defender channels + FIM configured
[ ] Custom rules loaded; manager restarted without errors
[ ] Smoke-test alerts visible under rule.groups:"ctm_capstone"
[ ] Snapshot taken on both VMs: "Day3-Wazuh-Ready"
```

Day 4 - The incident and your investigation

Goal: detect, investigate and document the intrusion. **Checkpoint:** a complete attack timeline and all alerts triaged.

4.1 Run the scenario (Red Cell only)

FILE

`05_Attack_Simulation / Run_Scenario.ps1`

Only the **Lab Engineer (Red Cell)** or the instructor runs this, inside the Windows VM, as Administrator. They do **not** read the script or share its output. Everyone else investigates blind, like a real SOC. First take a snapshot named **Day4-Before-Attack**.

DO THIS (Red Cell): `.\Run_Scenario.ps1` - it takes about 6 minutes and prints only "Stage X of 8 complete". Then wait 2-3 minutes for Wazuh to index events.

Plan B. If your live lab is broken (agent never connected, a VM died), use `06_Backup_Dataset\` instead. It contains the exact same incident as CSV and JSON files you can open in Excel and a text editor. Your investigation and report are identical.

4.2 How to investigate (the SOC method)

Work this loop for every alert. Record everything in `07_Team_Templates / 03_Alert_Triage_Tracker.xlsx` and `04_Investigation_Worksheet.docx`.

1. **Spot** - in Wazuh, search `rule.groups:"ctm_capstone"` and sort by time.
2. **Understand** - open the alert; read rule.description, rule.level, MITRE ID, and the event fields (account, command line, file).
3. **Confirm** - find the **same event in Event Viewer** using your custom views. This raw evidence is what goes in the report.
4. **Record** - add a row to the triage tracker: time, rule.id, Event ID, true/false positive, severity, MITRE, screenshot name.
5. **Connect** - place it on the timeline. Ask: what came before, what came after?

Take a screenshot of **every** confirmed alert (both the Wazuh view and the Event Viewer entry). Name them like `T03_4720_new_user.png`. Screenshots are your evidence and your presentation material.

4.3 What to look for (without spoiling it)

The intrusion moves through these ATT&CK stages. Find the evidence for each:

Stage	Question to answer	Where to look
Initial Access	How did they get in? Whose account?	4625, 4624; rule 100201
Discovery	What did they look at?	Sysmon 1 / 4688; rule 100206
Privilege / Persistence	New account? Auto-start? Task?	4720, 4732, 4698, Sysmon 13; rules 100202-100207
Execution	What code ran, and was it hidden?	4104, Sysmon 1; rule 100205
Collection / Exfil	What data was staged, changed, deleted, sent out?	FIM, Sysmon 11/22/3/26; rules 100208/100210
Tooling	Did AV catch anything?	Defender 1116/1117; rule 100209

Answer the 15 questions in the Investigation Worksheet. Decode the encoded PowerShell with **CyberChef** (From Base64) or by reading Event **4104**, which logs the decoded code.

> **Day 4 checkpoint**
> `[ ] Every ctm_capstone alert triaged in the tracker`
> `[ ] Each confirmed with an Event Viewer screenshot`
> `[ ] All 15 worksheet questions answered with evidence`
> `[ ] Attack timeline drafted (minute by minute)`
> `[ ] Each stage mapped to MITRE ATT&CK`

Day 5 - Report and present

Goal: deliver a professional incident report and a team presentation. **Checkpoint:** both complete and saved in the shared folder.

5.1 Write the incident report

FILE

`07_Team_Templates / 05_Incident_Report_Template.docx`

Fill every section. The executive summary must be readable by a non-technical manager. Your timeline and MITRE map come straight from Day 4. End with a clear **root cause** (hint: how strong was j.santos's password, and what rights did the account have?) and at least **five prioritised recommendations**.

5.2 Clean up and recover

FILE

`05_Attack_Simulation / Cleanup_Scenario.ps1`

After all evidence is collected, the Red Cell runs this to remove the backdoor account, scheduled task, Run key and dropped files, and to restore the finance data. In your report, describe this as your **containment, eradication and recovery**. (A real team would also reset the weak password and remove unnecessary admin rights - list that too.)

5.3 Build the presentation

FILE

`07_Team_Templates / 06_Final_Presentation_Outline.docx`

5.4 Peer evaluation

Each member completes `08_Assessment / Peer_Evaluation_Form.docx` and submits it privately to the instructor.

> **Day 5 checkpoint (final hand-in)**
> `[ ] Incident report complete`
> `[ ] Presentation delivered; every member spoke`
> `[ ] Triage tracker, worksheet, charter, stand-up log saved`
> `[ ] Lab cleaned up / recovered; final snapshot taken`
> `[ ] Peer evaluations submitted`

4. How you are graded

See `08_Assessment / Grading_Rubric.xlsx` for the full breakdown (100 points). Headline weights:

Area	Weight
Lab build & logging (Days 1-2)	15
Wazuh integration (Day 3)	15
Detection quality	10
Investigation & triage (Day 4)	20
Timeline & MITRE mapping	10
Incident report	15
Final presentation	10
Teamwork & documentation	5

5. Troubleshooting

Problem	Try this
VM won't start / 64-bit greyed out	Enable Intel VT-x / AMD-V in BIOS. Disable Hyper-V on the host if it conflicts.
VMs can't reach each other	Both need Adapter 2 = the **same** Host-only Adapter. Check IPs are both 192.168.56.x.
Scripts are blocked	Run `Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass` in that PowerShell window, as Administrator.
Wazuh agent not ACTIVE	Run `Preflight_Check.ps1 -WazuhIP <ip>`. Confirm ports 1514/1515 reachable; restart `WazuhSvc`; read `ossec.log`.
Agent version error	Agent version must be <= server version. Reinstall with the right `-Version`.
Custom rules not firing	Restart the manager. Check `/var/ossec/logs/ossec.log` for XML errors. Test with `wazuh-logtest`.
No Sysmon events	Confirm the **Sysmon64** service is running; re-run `Install_Sysmon.ps1`.
Dashboard won't load	Give the Wazuh VM more RAM (8 GB). Wait 2-3 minutes after boot. Use `https://` and accept the certificate.
Live lab broke on Day 4	Switch to `06_Backup_Dataset` and continue the investigation from the files.

Still stuck after 15 minutes? Post in your team channel with a screenshot of the error and what you already tried.

6. Glossary

Term	Plain meaning
SIEM	Security Information and Event Management - software that collects logs and raises alerts (Wazuh).
SOC	Security Operations Center - the team that watches for and responds to attacks (you).
Agent	Small program on a computer that sends its logs to the SIEM.
Sysmon	Microsoft tool that records detailed endpoint activity.
Event ID	A number Windows uses to label a type of event (e.g. 4625 = failed logon).
FIM	File Integrity Monitoring - alerts when watched files change.
Triage	Quickly deciding if an alert is real and how urgent it is.
True/False positive	A real detection vs. a harmless one.
Persistence	Techniques that let an attacker regain access after a reboot.
MITRE ATT&CK	A public catalogue of attacker techniques, each with an ID like T1110.
Snapshot	A saved state of a VM you can instantly roll back to.

7. Appendix - resource index

Everything below is in the single **CAPSTONE_RESOURCES** folder.

Folder	Used on	Contents
01_Guide	All	This guide (PDF)
02_Lab_Setup	Day 1	Download links, network plan, Preflight_Check.ps1
03_Windows_Logging	Day 2	Seed, audit, Sysmon scripts + config, custom views, cheat sheet
04_Wazuh	Day 3	Agent installer, ossec additions, local_rules.xml, dashboard queries
05_Attack_Simulation	Day 4	Scenario brief, Run_Scenario.ps1, Cleanup_Scenario.ps1
06_Backup_Dataset	Day 4 (Plan B)	Security/Sysmon CSV, Wazuh alerts JSON
07_Team_Templates	All	Charter, stand-ups, triage tracker, worksheet, report, presentation
08_Assessment	Day 5	Grading rubric, peer evaluation

End of guide - good luck, SOC team. Work the evidence, trust the timeline, and tell the story.